

Hello aspiring JEMIBB*,

Congratulations on passing the interview phase! You are one step away from becoming an integral part of our IT&BI team. To test your technical skills and your teamwork capabilities, we have devised a project that will allow you to put your data analysis and representation skills into practice.

Sentiment Analysis

Context

JEMIB is increasingly expanding its horizons. In particular, given the prestige of the IT&BI Area, we were contacted by Emma Martin, a French psychologist currently living in Italy and specializing in mass psychology. Currently, her research project aims to understand how sentiment on social networks has evolved over the last decade. Recently, she managed to obtain data from some posts that she only has in raw format (CSV) and which have already been classified. Her goal is to build dashboards that allow visualizing the current sentiment analysis. The latest available, but unclassified data are in a separate CSV, which will be provided to the algorithm every month.

Objective

Develop an algorithm that automatically assigns the sentiment related to the post and create dashboards, static or dynamic, that allow visualizing the main insights (historical trend, partition of various sentiments, etc.).

Dataset

The CSV file currently used by the psychologist - sentiment_analysis_train.CSV - and the one provided monthly - sentiment_analysis_test.CSV - have the following format:

- Year
- Month
- Day
- Time of Tweet
- Text
- Sentiment
- Platform

Project Phases

1. Data Analysis

- **Description:** Extract relevant and interesting insights from the dataset of classified posts to present to colleagues and the public. Every analysis and conclusion you draw must have a solid statistical justification, and everything must be written in a clear report.

2. Algorithm Development

- **Description:** Develop an algorithm capable of classifying the sentiment of the posts. In this phase, it is also important to evaluate the algorithm's results to support it.

3. Dashboard

- **Description:** Develop dashboards that visually show the results you obtained in point 1.
- **Optional:** If you are able to, create a small web page that gives the possibility to interact with the dashboards, for example with the possibility to set specific filters.

Tools and resources

We do not impose any constraints: feel free to use the tools you prefer. Within JEMIB, particularly in the IT & BI area, we use very diverse technologies and believe that the tool should be chosen based on the project's goals. Remember: JEMIB's motto is "LEARNING BY DOING". Do not hesitate to experiment with new technologies if you think they can bring value to the project!

NOTE. Should you use open-source software or version control software (like Github, Gitlab, etc.), do not forget to add your Buddy's and the Area Manager's email to your working team.

Additional details

To evaluate your teamwork, communications among yourselves during the project must take place entirely in a dedicated Telegram group. Inside, besides you, there will also be one of our associates, your Buddy, who will evaluate how you collaborate and whom you can contact if you have questions. Furthermore, on a date to be determined, you will have to present your work in a call, so be prepared to explain the choices you made and the reasons that led you to make them.

Delivery

Once completed, your material must be delivered by only one of you via email. The email must follow these guidelines:

- **RECIPIENT:** francesco.santamaria@jemib.it (CCing emilygrace.campanelli@jemib.it,

recruiter@jemib.it)

- **SUBJECT:** "TRIAL MEMBERS TASK | Sentiment Analytics"
- **ATTACHMENTS:**
 - A complete final report of the work done, including both the results obtained in point 1 and point 2.
 - Code used to develop the algorithm.
 - Images or links of the developed dashboards.

DEADLINE:

By and NO LATER THAN May 3, 2026

For any doubts, do not hesitate to contact us.

Good luck and good work!

The IT&BI Area of JEMIB